

Gaussian Location Family Optimal E-variable

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In this note, I will verify the Optimal E-variable of following testing problem:

$$\mathcal{H}_0 : \{N(\mu, \Sigma_p) | \mu \in \mathbb{R}^d\}, \quad \mathcal{H}_1 : \{N(0, \Sigma_q)\}.$$

We first see the simple case: $\Sigma_p = I_d, \Sigma_q = I_d + \Lambda$. (Λ is a diagonal matrix).

$$\mathcal{H}_0 : \{N(\mu, I_d) | \mu \in \mathbb{R}^d\}, \quad \mathcal{H}_1 : \{N(0, I_d + \Lambda)\}.$$

1 Problem Statement

At first glance, we could expect E-variable will be

$$S = \frac{p_{0,\Sigma}}{p_{0,I_d}}.$$

However, this fails to be an E-variable.

The Reverse Information projection of $\mathcal{H}_1$ to $\mathcal{H}_0$ is $p_{0,\Sigma}$ but, to be an E-variable, we need a RPr to $\text{conv}(\mathcal{H}_0)$. In this problem,

$$(\text{RPr to } \mathcal{H}_0) \neq (\text{RPr to } \text{conv}(\mathcal{H}_0)).$$

One known fact is, there exists exactly one distribution $p^* \in \text{conv}(\mathcal{H}_0)$ that q/p^* is an E-variable, and this distribution is RPr to $\text{conv}(\mathcal{H}_0)$.

2 Core Idea

[Hao+24] the simple case solved the problem when $\Sigma_p - \Sigma_q$ is positive semidefinite. In this case, the RPr is the likelihood ratio.

$$E_{GRO} = \frac{p_{0,\Sigma}}{p_{0,I_d}}.$$

$$(\text{RPr to } \mathcal{H}_0) = (\text{RPr to } \text{conv}(\mathcal{H}_0)).$$

[HG25] the general case solved the problem when $\Sigma_p - \Sigma_q$ is negative definite (they called 'Anti-simple' case). In this case, the RPr is $\mathcal{H}_1$ itself. Thus, there does not exist nontrivial E-variable.

$$E_{GRO} = 1.$$

Specifically, $\text{conv}(\mathcal{H}_0)$ contains $\mathcal{H}_1$.

However, in general there Σ_p and Σ_q does not have such a relationship. But, we can observe that for those two problems, the Reverse Information Projection was given by **Averaging Nulls for which the RPr to $\mathcal{H}_0$ fails to be an E-variable**. In the first case, there are no direction that RPr fails to be an E-variable. In the second case, RPr fails to be an E-variable in every directions.

3 RPr: The Saddle case

From the Core Idea, I propose the Growth Rate Optimal E-variable.

Theorem 1. For the testing problem $\Sigma_p = I_d, \Sigma_q = \Sigma = I_d + \Lambda$ where Λ is a diagonal matrix, and

$$\Lambda = \begin{pmatrix} \Lambda_- & 0 \\ 0 & -\Lambda_+ \end{pmatrix}$$

where Λ_-, Λ_+ are diagonal matrix with Λ_- having positive diagonal entries, Λ_+ having nonnegative diagonal entries. Each of dimension $\lambda \times \lambda$ and $(d - \lambda) \times (d - \lambda)$.

$$\mathcal{H}_0 : \{N(\mu, I_d) | \mu \in \mathbb{R}^d\}, \quad \mathcal{H}_1 : \{N(0, I_d + \Lambda)\}.$$

The Reverse Information Projection of $\mathcal{H}_1$ to $\mathcal{H}_0$ is given by

$$p^* = \text{Bayes Posterior of } p_{\mu, I_d} \text{ where } \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \mu_1 \sim N_\lambda(0, \Delta), \mu_2 = 0_{d-\lambda}.$$

$$\Delta = \Lambda_-.$$

Consequently, the Growth Rate Optimal E-variable is given by

$$E_{GRO} = \frac{q}{p^*} = \frac{1}{|I - \Lambda_+|^{1/2}} \exp \left(-\frac{1}{2} y_2^t \{ (I_{d-\lambda} - \Lambda_+)^{-1} - I_{d-\lambda} \} y_2 \right).$$

Proof. We show it by proving q/p^* become e-variable.

$$\begin{aligned} p^* &= \frac{1}{(2\pi)^{d/2}} \exp(-\frac{1}{2}|y|^2) \int \frac{1}{(2\pi)^{\lambda/2} |\Delta|^{1/2}} \exp \left(-\frac{1}{2} \mu_1^t \Delta^{-1} \mu_1 - \frac{1}{2} \mu_1^t \mu_1 + y_1^t \mu_1 \right) d\mu_1 \\ &= p_{0, I_d}(y) \frac{1}{|\Delta|^{1/2} |I_\lambda + \Delta^{-1}|^{1/2}} \exp \left(\frac{1}{2} y_1^t (I_\lambda + \Delta^{-1})^{-1} y_1 \right) \\ \frac{p_{\mu_0, I_d}(y)}{p^*(y)} &= |\Delta + I_\lambda|^{1/2} \exp \left(-\frac{1}{2} y_1^t (I_\lambda + \Delta^{-1})^{-1} y_1 \right) \frac{p_{\mu_0, I_d}(y)}{p_{0, I_d}(y)} \\ &= |\Delta + I_\lambda|^{1/2} \exp \left(-\frac{1}{2} y_1^t (I_\lambda + \Delta^{-1})^{-1} y_1 \right) \exp \left(-\frac{1}{2} \mu_0^t \mu_0 + \mu_0^t y \right) \\ \int \frac{qp_{\mu, I_d}}{p^*} &= \int \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} |\Delta + I_\lambda|^{1/2} \exp \left(-\frac{1}{2} y_1^t (I_\lambda + \Delta^{-1})^{-1} y_1 \right) \exp \left(-\frac{1}{2} \mu_0^t \mu_0 + \mu_0^t y \right) \exp \left(-\frac{1}{2} y^t \Sigma^{-1} y \right) \\ &= \frac{|\Delta + I_\lambda|^{1/2}}{|\Sigma|^{1/2}} \exp \left(-\frac{1}{2} y^t \left\{ \begin{pmatrix} (I_\lambda + \Delta^{-1})^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \Sigma^{-1} \right\} y + \mu_0^t y \right) \exp \left(-\frac{1}{2} \mu_0^t \mu_0 \right) \\ &= \frac{|\Delta + I_\lambda|^{1/2}}{|\Sigma|^{1/2}} \left| \left\{ \begin{pmatrix} (I_\lambda + \Delta^{-1})^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \Sigma^{-1} \right\} \right|^{-1/2} \exp \left(-\frac{1}{2} \mu_0^t \mu_0 \right) \exp \left(\frac{1}{2} \mu_0^t \left\{ \begin{pmatrix} (I_\lambda + \Delta^{-1})^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \Sigma^{-1} \right\}^{-1} \mu_0 \right) \\ &= \frac{|\Delta + I_\lambda|^{1/2}}{|\Sigma|^{1/2}} \left| \left\{ \begin{pmatrix} (I_\lambda + \Delta^{-1})^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \Sigma^{-1} \right\} \right|^{-1/2} \exp \left(-\frac{1}{2} \mu_0^t \left[I_d - \left\{ \begin{pmatrix} (I_\lambda + \Delta^{-1})^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \Sigma^{-1} \right\}^{-1} \right] \mu_0 \right) \end{aligned}$$

By Our choice $\Delta = \Lambda_-$,

$$\begin{aligned} \left[I_d - \left\{ \begin{pmatrix} (I_\lambda + \Delta^{-1})^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \Sigma^{-1} \right\}^{-1} \right] &= \left[I_d - \left\{ \begin{pmatrix} (I_\lambda + \Lambda_-^{-1})^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} (I_\lambda + \Lambda_-)^{-1} & 0 \\ 0 & (I_{d-\lambda} - \Lambda_+)^{-1} \end{pmatrix} \right\}^{-1} \right] \\ &= \left[I_d - \left\{ \begin{pmatrix} (I_\lambda + \Lambda_-^{-1})^{-1} + (I_\lambda + \Lambda_-)^{-1} & 0 \\ 0 & (I_{d-\lambda} - \Lambda_+^{-1})^{-1} \end{pmatrix} \right\}^{-1} \right] \\ &= \left[I_d - \left\{ \begin{pmatrix} I_\lambda & 0 \\ 0 & (I_{d-\lambda} - \Lambda_+^{-1})^{-1} \end{pmatrix} \right\}^{-1} \right] \\ &= \left[I_d - \begin{pmatrix} I_\lambda & 0 \\ 0 & I_{d-\lambda} - \Lambda_+^{-1} \end{pmatrix} \right] \\ &= \begin{pmatrix} 0 & 0 \\ 0 & \Lambda_+^{-1} \end{pmatrix} \end{aligned}$$

Thus,

$$\begin{aligned}
\int \frac{qp_{\mu, I_d}}{p^*} &= \frac{|\Delta + I_\lambda|^{1/2}}{|\Sigma|^{1/2}} \left| \left\{ \begin{pmatrix} (I_\lambda + \Delta^{-1})^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \Sigma^{-1} \right\} \right|^{-1/2} \exp \left(-\frac{1}{2} \mu_0^t \begin{pmatrix} 0 & 0 \\ 0 & \Lambda_+^{-1} \end{pmatrix} \mu_0 \right) \\
&= \frac{|\Delta + I_\lambda|^{1/2}}{|\Sigma|^{1/2}} \left| I_d - \begin{pmatrix} 0 & 0 \\ 0 & \Lambda_+^{-1} \end{pmatrix} \right|^{1/2} \exp \left(-\frac{1}{2} \mu_0^t \begin{pmatrix} 0 & 0 \\ 0 & \Lambda_+^{-1} \end{pmatrix} \mu_0 \right) \\
&= \frac{|\Lambda_- + I_\lambda|^{1/2}}{|I_\lambda + \Lambda_-|^{1/2} |I_{d-\lambda} - \Lambda_+|^{1/2}} |I_{d-\lambda} - \Lambda_+|^{1/2} \exp \left(-\frac{1}{2} \mu_0^t \begin{pmatrix} 0 & 0 \\ 0 & \Lambda_+^{-1} \end{pmatrix} \mu_0 \right) \\
&= \exp \left(-\frac{1}{2} \mu_0^t \begin{pmatrix} 0 & 0 \\ 0 & \Lambda_+^{-1} \end{pmatrix} \mu_0 \right) \leq 1.
\end{aligned}$$

□

Corollary 1. *The KL divergence has infimum*

$$\inf_{\mathbb{P} \in \text{conv}(\mathcal{H}_0)} KL(\mathbb{Q} \parallel \mathbb{P}) = -\frac{1}{2} \text{tr}(\Lambda_+) - \frac{1}{2} \log |I - \Lambda_+|.$$

Proof. By strong duality, the infimum of KL divergence is achieved at $\mathbb{P} = p^*$. Therefore,

$$\begin{aligned}
\inf_{\mathbb{P} \in \text{conv}(\mathcal{H}_0)} KL(\mathbb{Q} \parallel \mathbb{P}) &= \mathbb{E}_{N(0, \Sigma)} \log E_{GRO} \\
&= -\frac{1}{2} \log |I - \Lambda_+| - \frac{1}{2} \text{tr} \left(\{ (I_{d-\lambda} - \Lambda_+)^{-1} - I_{d-\lambda} \} \Sigma_{22} \right) \\
&= -\frac{1}{2} \log |I - \Lambda_+| - \frac{1}{2} \text{tr} \left(\{ (I_{d-\lambda} - \Lambda_+)^{-1} - I_{d-\lambda} \} (I_{d-\lambda} - \Lambda_+) \right) \\
&= -\frac{1}{2} \log |I - \Lambda_+| - \frac{1}{2} \text{tr}(\Lambda_+)
\end{aligned}$$

□

For the general hypothesis testing, we change variable. If Y follows the hypothesis,

$$\mathcal{H}_0 : \{N(\mu, \Sigma_p) | \mu \in \mathbb{R}^d\}, \quad \mathcal{H}_1 : \{N(0, \Sigma_q)\}.$$

then $X = \Sigma_p^{-1/2} Y$ follows $\{N(0, I_d)\}$ for null families, $N(0, \Sigma_p^{-1/2} \Sigma_q \Sigma_p^{-1/2})$ for alternative. Moreover, we can choose $\Sigma_p^{1/2}$ so that $\Sigma_p^{-1/2} \Sigma_q \Sigma_p^{-1/2}$ become a diagonal matrix, with diagonal entries in decreasing order. If

$$\Sigma_p^{-1/2} \Sigma_q \Sigma_p^{-1/2} = I + \begin{pmatrix} \Lambda_- & 0 \\ 0 & -\Lambda_+ \end{pmatrix}$$

the Growth Rate Optimal E-variable will be given by

$$E_{GRO}(x) = \frac{1}{|I - \Lambda_+|^{1/2}} \exp \left(-\frac{1}{2} x_2^t \{ (I_{d-\lambda} - \Lambda_+)^{-1} - I_{d-\lambda} \} x_2 \right).$$

thus,

$$E_{GRO}(y) = \frac{1}{|I - \Lambda_+|^{1/2}} \exp \left(-\frac{1}{2} \{ (\Sigma_p^{-1/2} y)_2 \}^t \{ (I_{d-\lambda} - \Lambda_+)^{-1} - I_{d-\lambda} \} \{ (\Sigma_p^{-1/2} y)_2 \} \right).$$

Corollary 2. *For general hypothesis testing*

$$\mathcal{H}_0 : \{N(\mu, \Sigma_p) | \mu \in \mathbb{R}^d\}, \quad \mathcal{H}_1 : \{N(0, \Sigma_q)\}.$$

the GRO E-variable is given by

$$E_{GRO}(y) = \frac{1}{|I - \Lambda_+|^{1/2}} \exp \left(-\frac{1}{2} \{ (\Sigma_p^{-1/2} y)_2 \}^t \{ (I_{d-\lambda} - \Lambda_+)^{-1} - I_{d-\lambda} \} \{ (\Sigma_p^{-1/2} y)_2 \} \right).$$

where $\Sigma_p^{1/2}$ satisfies

- (1) $\Sigma_p^{1/2} \Sigma_p^{1/2} = \Sigma_p$,
- (2) $\Sigma_p^{-1/2} \Sigma_q \Sigma_p^{-1/2}$ is a diagonal matrix, diagonals in decreasing order.

Moreover, the strong duality holds

$$\inf_{\mathbb{P} \in \text{conv}(\mathcal{H}_0)} KL(\mathbb{Q} \parallel \mathbb{P}) = \sup_E \mathbb{E}^{\mathbb{Q}}(\log E) = \frac{1}{2} \left\{ \sum_{i=1}^d \{ -(1 - \lambda_i)_+ - \log(1 - (1 - \lambda_i)_+) \} \right\},$$

where λ_i are eigenvalues of $\Sigma_p^{-1/2} \Sigma_q \Sigma_p^{-1/2}$.

4 Testing the Variance

4.1 Composite Alternative

Consider the testing problem

$$\mathcal{H}_0 : \{N(\mu, \Sigma_p) | \mu \in \mathbb{R}^d\}, \quad \mathcal{H}_1 : \{N(\mu, \Sigma_q) | \mu \in \mathbb{R}^d\}$$

Then we can propose following E-variables. Suppose we are given the sequence of observations $X_1, X_2, \dots, X_N$.

Universal Inference: since we know the GRO E-variable in the single alternative case, define $\hat{\mu}_j, j = 1, 2, \dots, N$ by

$$\hat{\mu}_j = \frac{n_0 q_0 + \sum_{i=1}^{j-1} x_i}{n_0 + j - 1},$$

which is a Bayes estimator of μ . Then

$$E_{UI} = \prod_{j=1}^N E_{GRO, \hat{\mu}_j}(X_j)$$

is an E-variable.

Another method is using the *GROW E-variable*. By the Perez et al ([Pér+23]), the GROW E-variable is the likelihood ratio of maximally invariant statistic, which is $(X_1 - \bar{X}, \dots, X_n - \bar{X})$, thus is conditional $\bar{X}$ distributions. ([Grü+25]).

Since the multivariate normal distribution is expressed as

$$p_{N(\mu, \Sigma)}(X_1, \dots, X_N) = \frac{1}{(\det(2\pi\Sigma))^{N/2}} \exp \left(-\frac{N}{2} (\bar{X} - \mu)^t \Sigma^{-1} (\bar{X} - \mu) - \frac{1}{2} \sum_{i=1}^N (X_i - \bar{X})^t \Sigma^{-1} (X_i - \bar{X}) \right),$$

$$p_{N(\mu, \Sigma)}(X_1 - \bar{X}, \dots, X_n - \bar{X}) = \frac{1}{(\det(2\pi\Sigma))^{(N-1)/2} N^{d/2}} \exp \left(-\frac{1}{2} \sum_{i=1}^N (X_i - \bar{X})^t \Sigma^{-1} (X_i - \bar{X}) \right)$$

$$E_{GROW} = \frac{q(\cdot | \bar{X})}{p(\cdot | \bar{X})} = \left(\frac{\det \Sigma_p}{\det \Sigma_q} \right)^{(N-1)/2} \exp \left(-\frac{1}{2} \sum_{i=1}^N (X_i - \bar{X})^t (\Sigma_q^{-1} - \Sigma_p^{-1}) (X_i - \bar{X}) \right).$$

4.2 Composite Null

E-value for the hypothesis testing problem

$$\mathcal{H}_0 : \{N(\mu, \Sigma_p) | \mu \in \mathbb{R}^d\}, \quad \mathcal{H}_1 : \{N(\mu, \Sigma_q)\}$$

is also an E-variable for the testing problem

$$\mathcal{H}_0 : \{N(\mu, \Sigma) | \mu \in \mathbb{R}^d, \Sigma \succeq \Sigma_p\}, \quad \mathcal{H}_1 : \{N(\mu, \Sigma_q)\}.$$

This is because, $\mathbb{E}_{N(\mu, \Sigma_p)}[E] \leq 1$ implies $\mathbb{E}_{N(\mu_0, \Sigma_0)}[E] \leq 1$ for $\Sigma_0 \succeq \Sigma_p$ and arbitrary μ_0 , because $\mu \sim N(\mu_0, \Sigma_0 - \Sigma_p)$ gives the inequality.

5 Geometric Interpretation

In Theorem 1, we obtained the reverse information projection by mixing null distributions in a way that stretches the variance. To explain, [Hao+24] showed that if $\Sigma_p - \Sigma_q$ is positive definite, then we do not need a mixture. My result additionally proved that, for the subspace on which $\Sigma_p - \Sigma_q$ is negative definite, we can mix the null distributions to **inflate** the variance. From this perspective, Theorem 1 suggests mixing over $N(0, \Lambda_-)$ so that the mixture distribution satisfies $\Sigma_{\text{mixture}} - \Sigma_q \succeq 0$.

This intuition can be interpreted from an information-geometric perspective. In information geometry, assuming an exponential family $p(x|\theta) = \exp(x^t \theta - \psi(\theta))$, the information matrix $I(\theta) = \partial_\theta^2 \psi(\theta)$ defines a Riemannian metric on the natural parameter space θ . Mixing distributions stretches the variance, and we can therefore suspect that the variance, and hence the corresponding metric, should satisfy a domination relation.

Moreover, we can derive some properties of the numeraire from this perspective. Assume the following hypothesis testing problem:

$$H_0 : \mathbb{P} \in \mathcal{P} = \{\mathbb{P}_\theta = \exp(\theta^t x - \psi(\theta) + C(x)) | \theta \in \Theta\}, \quad H_1 : \mathbb{Q}.$$

5.1 E-value is strict on its support

Assume that the reverse information projection of $\mathbb{Q}$ onto $\text{conv}(\mathcal{P})$ is a mixture over a d -dimensional submanifold $\mathcal{M}$ of $\mathcal{P}$, i.e., a curved exponential family. Thus, the reverse information projection is of the form

$$\hat{\mathbb{P}} = \int_{\mathcal{M}} \phi(u) \mathbb{P}_{\theta(u)} du, \quad \int_{\mathcal{M}} \phi(u) du = 1.$$

Then, we can show that

$$\mathbb{E}^{\mathbb{P}_{\theta(u)}} \left[\frac{\mathbb{Q}}{\hat{\mathbb{P}}} \right] = 1.$$

To show this, let φ be supported on $\theta(u) \in U \subset \mathcal{M}$ and satisfy $\int_U \varphi(u) du = 1$. Then, for sufficiently small $\alpha \in (-\epsilon, \epsilon)$,

$$(1 - \alpha) \int_{\mathcal{M}} \phi(u) \mathbb{P}_{\theta(u)} du + \alpha \int_{\mathcal{M}} \varphi(u) \mathbb{P}_{\theta(u)} du$$

is a probability distribution and an element of $\text{conv}(\mathcal{P})$.

Since

$$KL \left(\mathbb{Q} \left\| (1 - \alpha) \int_{\mathcal{M}} \phi(u) \mathbb{P}_{\theta(u)} du + \alpha \int_{\mathcal{M}} \varphi(u) \mathbb{P}_{\theta(u)} du \right. \right) \geq KL(\mathbb{Q} \| \hat{\mathbb{P}}),$$

taking the derivative with respect to α at $\alpha = 0$ gives

$$\int \frac{\mathbb{Q}}{\hat{\mathbb{P}}} \left(\int_{\mathcal{M}} \varphi(u) \mathbb{P}_{\theta(u)} - \hat{\mathbb{P}} \right) = 0,$$

and hence

$$\int_U \left\{ \int \frac{\mathbb{Q}(x)}{\hat{\mathbb{P}}(x)} \mathbb{P}_{\theta(u)}(x) \right\} \varphi(u) = 1.$$

Since this holds for every bump function φ , we obtain

$$\int \frac{\mathbb{Q}}{\hat{\mathbb{P}}} \mathbb{P}_{\theta(u)} = 1$$

for all $u \in U$.

Remark 1. *This strictness property holds not only for the reverse information projection, but also for a distribution that minimizes the KL divergence over a d -dimensional curved exponential mixture. The proof technique is exactly the same.*

5.2 First Order Condition

Using the derivative formula

$$\frac{d}{d\theta} \mathbb{P}_{\theta} = \mathbb{P}_{\theta} \cdot (x - \mu(\theta)),$$

we obtain

$$\int x \cdot \frac{\mathbb{Q}}{\hat{\mathbb{P}}} \mathbb{P}_{\theta(u)}(x) = \mu(\theta(u)) =: \mu(u).$$

By multiplying both sides by $\phi(u)$ and integrating over u , we obtain

$$\mathbb{E}^{\mathbb{Q}}[X] = \int x \cdot \mathbb{Q} = \int_{\mathcal{M}} \phi(u) \mu(u).$$

5.3 Second Order Condition

Similarly, we obtain

$$\int \frac{\mathbb{Q}}{\hat{\mathbb{P}}} \mathbb{P}_{\theta(u)} [(x - \mu)(x - \mu)^t] \preceq \Sigma_{\theta(u)}.$$

Moreover, equality holds along the tangent vectors of $\theta(u)$.

5.4 Quadratic Condition on Mixtures

The original numeraire problem is the following optimization problem:

$$\inf_{\mathbb{P} \in \text{conv}(\mathcal{H}_0)} KL(\mathbb{Q} \parallel \mathbb{P}).$$

Motivated by the first- and second-order conditions above, we may consider the constrained optimization problem

$$\inf_{\mathbb{P} \in \text{conv}(\mathcal{H}_0)} KL(\mathbb{Q} \parallel \mathbb{P}) \quad \text{subject to } \mu_{\mathbb{P}} = \mu_{\mathbb{Q}}, \quad \Sigma_{\mathbb{Q}} \preceq \Sigma_{\mathbb{P}}.$$

This problem has a natural convex-optimization structure. First, $\text{conv}(\mathcal{H}_0)$ is convex by construction. Second, $KL(\mathbb{Q} \parallel \mathbb{P})$ is convex as a function of $\mathbb{P}$. Finally, once the mean constraint $\mu_{\mathbb{P}} = \mu_{\mathbb{Q}} =: \mu$ is imposed, the covariance constraint can be written as

$$\int (x - \mu)(x - \mu)^t d\mathbb{P}(x) \succeq \Sigma_{\mathbb{Q}}.$$

which is affine in $\mathbb{P}$. Therefore, both constraints are affine moment constraints on the probability measure $\mathbb{P}$, while the objective is convex.

Let $\lambda \in \mathbb{R}^d$ denote the Lagrange multiplier associated with the mean constraint, and let $\Lambda \succeq 0$ denote the matrix-valued multiplier associated with the covariance constraint. Writing the covariance constraint as

$$\Sigma_{\mathbb{Q}} - \Sigma_{\mathbb{P}} \preceq 0,$$

the Lagrangian is

$$L(\mathbb{P}, \lambda, \Lambda) = KL(\mathbb{Q} \parallel \mathbb{P}) + \lambda^t(\mu_{\mathbb{P}} - \mu_{\mathbb{Q}}) + \langle \Lambda, \Sigma_{\mathbb{Q}} - \Sigma_{\mathbb{P}} \rangle,$$

Although the optimization variable $\mathbb{P}$ belongs to an infinite-dimensional space of probability measures, the constraints themselves are finite-dimensional. The mean constraint takes values in $\mathbb{R}^d$, whereas the covariance constraint takes values in the finite-dimensional cone of positive semidefinite matrices. Consequently, the associated dual variables are also finite-dimensional.

A natural analogue of Slater's condition is the existence of some $\bar{\mathbb{P}} \in \text{conv}(\mathcal{H}_0)$ such that

$$\mu_{\bar{\mathbb{P}}} = \mu_{\mathbb{Q}}, \quad \Sigma_{\bar{\mathbb{P}}} \succ \Sigma_{\mathbb{Q}},$$

together with

$$KL(\mathbb{Q} \parallel \bar{\mathbb{P}}) < \infty.$$

The corresponding KKT conditions will become

$$\begin{aligned} \mu_{\mathbb{P}^*} &= \mu_{\mathbb{Q}}, \quad \Sigma_{\mathbb{P}^*} \succeq \Sigma_{\mathbb{Q}}, \\ \Lambda^* &\succeq 0, \end{aligned}$$

and complementary slackness,

$$\langle \Lambda^*, \Sigma_{\mathbb{P}^*} - \Sigma_{\mathbb{Q}} \rangle = 0$$

The complementary slackness condition has a natural geometric interpretation. The matrix Λ^* can assign positive weight only to directions in which the covariance-domination constraint is active. Thus, Λ^* identifies the tangent directions along which the metric of the optimal mixture is just sufficient to dominate the metric associated with $\mathbb{Q}$.

Therefore, the covariance-inflation phenomenon in Theorem 1 can be interpreted as a primal-dual convex optimization problem. The primal problem searches over mixtures of null distributions that preserve the mean and dominate the covariance of $\mathbb{Q}$, while minimizing the reverse KL divergence. The dual matrix Λ^* characterizes the directions in which this covariance, or metric, domination constraint is binding.

Appendix A. Testing Mean

Testing Mean is nothing new: for multivariate case, I showed the reverse information projection to convex hull is simply the MLE distribution.

Consider the testing problem:

$$\mathcal{H}_0 : \{N(0, \Sigma) | \Sigma \in S^+\}, \quad \mathcal{H}_1 : \{q = N(\mu_q, \Sigma_q)\}.$$

The KL divergence is given by

$$\text{KL}(N(\mu_q, \Sigma_q) \| N(0, \Sigma)) = \frac{1}{2} \left[\text{tr}(\Sigma^{-1} \Sigma_q) - d + \mu_q^t \Sigma^{-1} \mu_q + \ln \frac{\det \Sigma}{\det \Sigma_q} \right].$$

Using the matrix differential, we could calculate:

$$\begin{aligned} d(\text{tr}(\Sigma^{-1} \Sigma_q)) &= -\text{tr}(\Sigma^{-1} \Sigma_q \Sigma^{-1} (d\Sigma)), \quad \frac{d}{d\Sigma} \text{tr}(\Sigma^{-1} \Sigma_q) = -\Sigma^{-1} \Sigma_q \Sigma^{-1}, \\ d(\mu^t \Sigma^{-1} \mu) &= -\mu^t \Sigma^{-1} d\Sigma \Sigma^{-1} \mu = -\text{tr}(\Sigma^{-1} \mu \mu^t \Sigma^{-1} d\Sigma), \quad \frac{d}{d\Sigma} (\mu^t \Sigma^{-1} \mu) = -\Sigma^{-1} \mu \mu^t \Sigma^{-1}, \\ d(\ln \det \Sigma) &= \text{tr}(\Sigma^{-1} d\Sigma), \quad \frac{d}{d\Sigma} \ln \det(\Sigma) = \Sigma^{-1}, \end{aligned}$$

Thus,

$$\frac{d}{d\Sigma} \text{KL}(N(\mu_q, \Sigma_q) \| N(0, \Sigma)) = \frac{1}{2} [-\Sigma^{-1} \Sigma_q \Sigma^{-1} - \Sigma^{-1} \mu_q \mu_q^t \Sigma^{-1} + \Sigma^{-1}],$$

which is zero at

$$\Sigma = \Sigma_q + \mu_q \mu_q^t,$$

thus, the Reverse information projection to $\mathcal{H}_0$ is $N(0, \Sigma_q + \mu_q \mu_q^t)$.

Remark 2. This coincides with 1-dimensional case, calculated from [Grü+25].

Next, we verify q/p^* is indeed an E-variable, thus optimal E-variable by calculating hessian. [Grü+25] gave a sufficient condition so that q/p^* become E-variable.

$\mathcal{H}_0$ forms an exponential family with natural coordinate system $\theta^{ii} = -\frac{1}{2}(\Sigma^{-1})_{ii}$, $\theta^{ij} = -(\Sigma^{-1})_{ij}$ with sufficient statistic $F_{ij}(x) = x_i x_j$, $1 \leq i < j \leq d$. Also write $\theta = -\frac{1}{2}\Sigma^{-1}$, $F(x) = xx^t$. Then

$$\begin{aligned} p &= p_{N(0, \Sigma_q + \mu_q \mu_q^t)} \exp \left(\sum_{1 \leq i < j \leq d} \theta^{ij} F_{ij}(x) \right) \exp(-\psi_p(\theta)), \\ q &= q_{N(\mu, \Sigma_q)} \exp \left(\sum_{1 \leq i < j \leq d} \theta^{ij} F_{ij}(x) \right) \exp(-\psi_q(\theta)). \end{aligned}$$

Calculating ψ_p, ψ_q gives,

$$\begin{aligned} \exp(\psi_p(\theta)) &= \int \frac{1}{\sqrt{\det(2\pi(\Sigma_q + \mu_q \mu_q^t))}} \exp \left(-\frac{1}{2} x^t (\Sigma_q + \mu_q \mu_q^t)^{-1} x + \text{tr}(\theta F(x)) \right) \\ &= \int \frac{1}{\sqrt{\det(2\pi(\Sigma_q + \mu_q \mu_q^t))}} \exp \left(-\frac{1}{2} x^t (\Sigma_q + \mu_q \mu_q^t)^{-1} x + x^t \theta x \right) \\ &= \frac{1}{\sqrt{\det(2\pi(\Sigma_q + \mu_q \mu_q^t))}} \frac{1}{\sqrt{\det(2\pi((\Sigma_q + \mu_q \mu_q^t)^{-1} - 2\theta))}} \\ &= \frac{1}{\sqrt{\det(I - 2\theta(\Sigma_q + \mu_q \mu_q^t))}}. \\ \exp(\psi_q(\theta)) &= \int \frac{1}{\sqrt{\det(2\pi\Sigma_q)}} \exp \left(-\frac{1}{2} (x - \mu_q)^t \Sigma_q^{-1} (x - \mu_q) + \text{tr}(\theta F(x)) \right) \\ &= \frac{1}{\sqrt{\det(I - 2\theta\Sigma_q)}} \exp \left(\frac{1}{2} \mu_q^t \{ \Sigma_q^{-1} (\Sigma_q^{-1} - 2\theta)^{-1} \Sigma_q^{-1} - \Sigma_q^{-1} \} \mu_q \right). \end{aligned}$$

Thus,

$$\psi_p(\theta) = -\frac{1}{2} \log \det(I - 2\theta(\Sigma_q + \mu_q \mu_q^t)),$$

$$\psi_q(\theta) = -\frac{1}{2} \log \det(I - 2\theta \Sigma_q) + \frac{1}{2} \mu_q^t \{ \Sigma_q^{-1} (\Sigma_q^{-1} - 2\theta)^{-1} \Sigma_q^{-1} - \Sigma_q^{-1} \} \mu_q.$$

For $f(\theta) = -1/2 \log \det(I - 2\theta \Sigma)$ and $g(\theta) = 1/2 \mu^t \{ \Sigma^{-1} (\Sigma^{-1} - 2\theta)^{-1} \Sigma^{-1} \} \mu$,

$$df = (I - 2\theta \Sigma)^{-1} d\theta \Sigma = \text{tr}(\Sigma(I - 2\theta \Sigma^{-1})^{-1} d\theta).$$

$$dg = \text{tr}((\Sigma^{-1} - 2\theta)^{-1} \Sigma^{-1} \mu \mu^t \Sigma^{-1} (\Sigma^{-1} - 2\theta)^{-1} d\theta).$$

$$\begin{aligned} f''(\theta)[d\theta, d\theta'] &= \frac{1}{2} d'[\text{tr}(\Sigma(I - 2\theta \Sigma)^{-1} d\theta)] \\ &= \frac{1}{2} \text{tr}[\Sigma \cdot (-1)(I - 2\theta \Sigma)^{-1} (-2d\theta' \Sigma)(I - 2\theta \Sigma)^{-1} d\theta] \\ &= \text{tr}[\Sigma(I - 2\theta \Sigma)^{-1} d\theta' \Sigma(I - 2\theta \Sigma)^{-1} d\theta] \\ &= \text{vec}(d\theta')^t [(I - 2\Sigma\theta)^{-1} \Sigma \otimes \Sigma(I - 2\theta \Sigma)^{-1}] \text{vec}(d\theta). \end{aligned}$$

$$\begin{aligned} g''(\theta)[d\theta, d\theta'] &= \frac{1}{2} d'[\text{tr}((\Sigma^{-1} - 2\theta)^{-1} \Sigma^{-1} \mu \mu^t \Sigma^{-1} (\Sigma^{-1} - 2\theta)^{-1} d\theta)] \\ &= \frac{1}{2} \text{tr}[-(I - 2\Sigma\theta)^{-1} (-2\Sigma d\theta') (I - 2\Sigma\theta)^{-1} \mu \mu^t (I - 2\theta \Sigma)^{-1} d\theta] \\ &\quad - (I - 2\Sigma\theta)^{-1} \mu \mu^t (I - 2\theta \Sigma)^{-1} (-2d\theta' \Sigma) (I - 2\theta \Sigma)^{-1} d\theta] \\ &= \text{tr}[(I - 2\Sigma\theta)^{-1} \Sigma d\theta' (I - 2\Sigma\theta)^{-1} \mu \mu^t (I - 2\theta \Sigma)^{-1} d\theta] \\ &\quad + (I - 2\Sigma\theta)^{-1} \mu \mu^t (I - 2\theta \Sigma)^{-1} d\theta' \Sigma (I - 2\theta \Sigma)^{-1} d\theta] \\ &= \text{vec}(d\theta')^t [\Sigma(I - 2\theta \Sigma)^{-1} \otimes (I - 2\Sigma\theta)^{-1} \mu \mu^t (I - 2\theta \Sigma)^{-1}] \text{vec}(d\theta) \\ &\quad + \text{vec}(d\theta')^t [(I - 2\Sigma\theta)^{-1} \mu \mu^t (I - 2\theta \Sigma)^{-1} \otimes \Sigma(I - 2\theta \Sigma)^{-1}] \text{vec}(d\theta). \end{aligned}$$

Thus, the Hessian of ψ_p, ψ_q , calculated at $\theta = 0$ is

$$\mathcal{H}_p = (\Sigma_q + \mu_q \mu_q^t) \otimes (\Sigma_q + \mu_q \mu_q^t).$$

$$\mathcal{H}_q = (\Sigma_q \otimes \Sigma_q) + (\Sigma_q \otimes \mu_q \mu_q^t) + (\mu_q \mu_q^t \otimes \Sigma_q).$$

Thus, $\mathcal{H}_p \geq \mathcal{H}_q$, meaning that this problem reduces to the simple case.

References

- [Pér+23] Muriel Felipe Pérez-Ortiz et al. *E-Statistics, Group Invariance and Anytime Valid Testing*. 2023. arXiv: 2208.07610 [math.ST]. URL: <https://arxiv.org/abs/2208.07610>.
- [Hao+24] Yunda Hao et al. *E-values for k-Sample Tests With Exponential Families*. 2024. arXiv: 2303.00471 [stat.ME]. URL: <https://arxiv.org/abs/2303.00471>.
- [Grü+25] Peter Grünwald et al. *Optimal E-Values for Exponential Families: the Simple Case*. 2025. arXiv: 2404.19465 [stat.ME]. URL: <https://arxiv.org/abs/2404.19465>.
- [HG25] Yunda Hao and Peter Grünwald. *E-Values for Exponential Families: the General Case*. 2025. arXiv: 2409.11134 [stat.ME]. URL: <https://arxiv.org/abs/2409.11134>.